GitHub Repository Details

Technical Documentation • Local System Setup • Execution Reference

OFFICIAL DOCUMENTATION

1. REPOSITORY SPECIFICATION & HYPERLINK

DIRECT REMOTE REPOSITORY URL (CLICK TO OPEN)

https://github.com/RohithBheemini/IDS-Windmill-Project — Open Repository →

PROJECT	OWNER	DEFAULT BRANCH	PRIMARY STACK
IDS-Windmill-Project	RohithBheemini	main	R (v4.0.0+) / Shiny

2. TAKING THE PROJECT INTO YOUR OWN SYSTEM (SETUP GUIDE)

Step 1: Clone Repository

```
# Clone repository via Git
git clone https://github.com/RohithBheemini/IDS-Windmill-Project.git

# Move into root directory
cd IDS-Windmill-Project
```

Step 2: Install R Package Dependencies

```
# Run in terminal / PowerShell once:
Rscript -e "install.packages(c('data.table',
'zoo','ranger','xgboost','pROC','lime',
'shiny','bslib','bsicons','DT','ggplot2',
'scales','rmarkdown','knitr'),
repos='https://cloud.r-project.org')"
```

3. RUNNING THE APPLICATION & WORKFLOWS (MAIN.R)

TARGET ACTION	CLI COMMAND	DESCRIPTION / OUTPUT
Launch Dashboard	`Rscript main.R --app`	Opens interactive Shiny UI at `http://127.0.0.1:3838`
Run Terminal Demo	`Rscript main.R --demo`	Simulates Event 72 breakdown, 14-day lead time & XAI output
Full Demo + App	`Rscript main.R`	Executes live CLI demo and automatically launches browser app
Retrain Models	`Rscript main.R --train`	Re-fits GLM, Random Forest, and XGBoost models (0.67s training)
Re-clean Telemetry	`Rscript main.R --clean`	Re-engineers physics features (ΔT, 6h rolling, gear slip)

4. CORE DELIVERABLES & FILE HIERARCHY

```
IDS-Windmill-Project/
├── main.R               # Unified controller script
├── shiny_app/
│   └── app.R            # 5-tab Shiny web dashboard
├── data_clean/
│   ├── clean_telemetry.rds # Cleaned SCADA dataset (109 MB)
│   └── clean_events.csv    # 44 anomaly & 51 normal events
├── models/
│   ├── xgb_model.rds    # XGBoost classifier (0.83 AUC)
│   ├── rf_model.rds     # Random Forest benchmark
│   ├── glm_model.rds    # Logistic Regression baseline
│   ├── care_earliness.rds # 8-34 days lead-time metrics
│   └── xai_global_shap.rds# TreeSHAP feature attributions
├── wind_turbine_ids_report.html # Full technical report
├── presentation_wind_turbine_ids.pptx # 12-slide deck
└── README.md            # Project documentation
```

COMPONENT	ROLE IN PROJECT
`main.R`	Single entry point for demo, dashboard, training, and pipeline.
`shiny_app/app.R`	Decision support app: telemetry studio, benchmarks, XAI, ROI.
`report.html`	Comprehensive technical documentation with lecturer Q&A.
`presentation.pptx`	16:9 widescreen presentation deck (titles 32–46pt, bright white).
`dataset_details.md`	Microscopic channel catalog and physical sensor dictionary.

Zero Cloud Dependencies: All telemetry and models are local `.rds` files. The application runs 100% offline without external services or credentials.